



ELSEVIER

Available online at www.sciencedirect.com

SCIENCE @ DIRECT®

Expo. Math. 23 (2005) 233–254

EXPOSITIONES
MATHEMATICAE

www.elsevier.de/exmath

Singularities of plane algebraic curves

Jonathan A. Hillman*

School of Mathematics and Statistics, The University of Sydney, Sydney, NSW 2006, Australia

Received 25 August 2004

Abstract

We give an exposition of some of the basic results on singularities of plane algebraic curves, in terms of polynomials and formal power series.

© 2005 Elsevier GmbH. All rights reserved.

MSC 2000: primary 14H20; secondary 32Sxx

Keywords: Conductor; Gauß–Manin connection; Isolated singularity; Milnor number; Nullstellensatz; Plane curve; Puiseux series

1. Introduction

A nonconstant polynomial $f \in \mathbb{C}[X, Y]$ defines a plane curve $V(f) = \{(x, y) \in \mathbb{C}^2 \mid f(x, y) = 0\}$. If f is square-free there are only finitely many points $P \in V(f)$ such that $f_X(P) = f_Y(P) = 0$, where f_X and f_Y are the partial derivatives of f . This article is an account of the basic properties of such singularities of plane curves, in terms of elementary commutative algebra. It began as an attempt to understand the work of Morisita and others on analogies between algebraic number theory and knot theory [28]. Although we have not pursued such analogies here, the possibility of transposing arguments into number theory has lead to our emphasizing the commutative-algebraic point of view. The geometric and topological aspects of plane curves and their singularities are treated in much greater detail in the books [10, 15, 27].

* Tel.: +61 2 9351 3049; fax: +61 2 9351 4534.

E-mail address: jonh@maths.usyd.edu.au.

In Section 2, we prove the weak Nullstellensatz, and show that an irreducible plane curve $V(f) \subset \mathbb{C}^2$ is smooth if and only if its coordinate ring $\mathbb{C}[X, Y]/(f)$ is integrally closed. In Section 3, we show that the formal power series ring $\mathbb{C}[[X, Y]]$ is a 2-dimensional factorial domain and establish the Weierstraß Division and Preparation Theorems. We also show that every square-free formal power series is equivalent to a polynomial after a formal change of coordinates. In Section 4, we pass to $A = \mathbb{C}[[X, Y]]/(f)$, the completion of the coordinate ring of $V(f)$ at 0, where f is a square-free distinguished polynomial in Y , and describe the method of Puiseux expansions. The next two sections consider numerical invariants of singularities. In Section 5, we show that the Milnor number $\mu(f) = \dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/(f_X, f_Y)$ is finite if and only if f is square-free, and we compute this for f a weighted homogeneous polynomial. In Section 6, we show that the codimension δ_A of A in its integral closure $\bar{A}$ equals the codimension of the conductor $\mathcal{C}$ of $\bar{A}$ into A in A , and that $\delta_A = (\mu(f) + 1 - r)/2$, where r is the number of irreducible factors of f . These invariants are used in Section 7 to outline how plane curve singularities may be resolved by iterated quadratic transforms. The final sections are devoted to calculating the monodromy of the local Gauß–Manin connection in the weighted homogeneous case.

Although all the results described here are well-known, we believe that some of the arguments may be new. The only algebraic result that we use which is not in the text [4] is the fact that the integral closure of a complete discrete valuation ring in a finite extension of its fields of fractions is again a complete discrete valuation ring, for which we refer to Serre [36].

2. Algebraic curves

If f is a nonconstant polynomial the irreducible components of $V(f)$ correspond to the irreducible factors of f . The points of $V(f)$ correspond to the maximal ideals of the coordinate ring $O_{V(f)} = \mathbb{C}[X, Y]/(f)$, by the Nullstellensatz. We shall give a simple ad hoc argument for the 2-variable case, which extends readily to a proof of the (weak) Nullstellensatz for polynomial rings in many variables over algebraically closed fields. (The arguments for Theorems 1 and 4 are based on ones given in [19].)

Theorem 1. *Let M be a maximal ideal in $\mathbb{C}[X, Y]$. Then $M = (X - \alpha, Y - \beta)$ for some $\alpha, \beta \in \mathbb{C}$.*

Proof. If $M \cap \mathbb{C}[X] = 0$ then $\mathbb{C}[X]$ embeds in the field $K = \mathbb{C}[X, Y]/M$, and so the projection extends to a homomorphism from $\mathbb{C}(X)[Y]$ to K . Hence the kernel $M_0 = M \cap \mathbb{C}(X)[Y]$ is a proper maximal ideal in the PID $\mathbb{C}(X)[Y]$, and so is generated by f/d say, where $f \in M$ has positive degree in Y and $d \in \mathbb{C}[X] - \{0\}$. Then f divides dM in $\mathbb{C}[X, Y]$ and so f and M generate the same (maximal) ideal in $\mathbb{C}[X, Y][1/d]$. Let $h(X)$ be the coefficient of the leading term of f , and let $\gamma \in \mathbb{C}$ be a point at which neither d nor h is 0. Since $X - \gamma$ is a nonzero element of the field K and $K = \mathbb{C}[X, Y][1/d]/(f)$ there are $g, k \in \mathbb{C}[X, Y][1/d]$ such that $(X - \gamma)g - 1 = fk$. Let $\rho : \mathbb{C}[X, Y][1/d] \rightarrow \mathbb{C}[Y]$ be the epimorphism with kernel $(X - \gamma)$. Then $\rho(f)$ is a unit in $\mathbb{C}[Y]$, and so $\rho(h) = 0$, contrary to hypothesis. Therefore $M \cap \mathbb{C}[X]$ is a nonzero prime ideal, and so is generated by $X - \alpha$, for some $\alpha \in \mathbb{C}$. Similarly M contains $Y - \beta$ for some $\beta \in \mathbb{C}$, and so $M = (X - \alpha, Y - \beta)$. $\square$

This theorem can be generalized as follows.

Let R be a domain in which the intersection of the maximal ideals is 0, and let M be a maximal ideal in $R[X]$. Then $M \cap R$ is a maximal ideal in R , and the intersection of all the maximal ideals of $R[X]$ is 0.

In the above argument we replace γ by a maximal ideal n of R which does not contain dh , $X - \gamma$ by a nonzero element of n and ρ by the projection of $R[1/d, Y]$ onto $(R/n)[Y]$. (Some hypothesis on R is necessary, for if $R = \mathbb{Z}_{(p)}$ is the ring of rationals with denominator prime to p and $f = 1 - pX$ then $R \cap (f) = 0$ while $R[X]/(f) = \mathbb{Q}$, so (f) is maximal in $R[X]$.) This result may be applied inductively to show that if K is an algebraically closed field the maximal ideals of $K[X_1, \dots, X_n]$ are all of the form $(X_1 - \kappa_1, \dots, X_n - \kappa_n)$. (This is the “Weak Nullstellensatz”.)

Theorem 2. *Let $f, g \in \mathbb{C}[X, Y]$ have no common factors. Then*

$$|V(f) \cap V(g)| \leq \dim_{\mathbb{C}} \mathbb{C}[X, Y]/(f, g) < \infty.$$

Proof. The images of f and g in $\mathbb{C}(X)[Y]$ have no common factor there, since $\mathbb{C}[X, Y]$ is a UFD. Hence there are $m, n \in \mathbb{C}(X)[Y]$ such that $mf + ng = 1$, by the Euclidean algorithm. Clearing denominators, there are polynomials $M, N \in \mathbb{C}[X, Y]$ and a monic polynomial $h \in \mathbb{C}[X]$ such that $Mf + Ng = h$. Hence the first coordinate of any point of $V(f) \cap V(g)$ is a root of h , and for each such root α the number of points in $V(f) \cap V(g)$ with first coordinate α is bounded by $\dim_{\mathbb{C}} \mathbb{C}[X, Y]/(X - \alpha, f, g)$. Since $X - \alpha$ does not divide both f and g this dimension is finite, and so $V(f) \cap V(g)$ is finite.

More precisely, let $h = \prod_{i=1}^r (X - \alpha_i)^{m(i)}$ be the factorization of h . Let $A_i = \mathbb{C}[X]/((X - \alpha_i)^{m(i)})$ and let f_i, g_i be the images of f, g in $A_i[Y] = \mathbb{C}[X, Y]/((X - \alpha_i)^{m(i)})$, for all $i \leq r$. Then $\dim_{\mathbb{C}} \mathbb{C}[X, Y]/(X - \alpha_i, f, g) \leq \dim_{\mathbb{C}} A_i[Y]/(f_i, g_i) < \infty$, for all $i \leq r$. As $\mathbb{C}[X, Y]/(f, g) \cong \bigoplus_{i=1}^r A_i[Y]/(f_i, g_i)$ we have $|V(f) \cap V(g)| \leq \dim_{\mathbb{C}} \mathbb{C}[X, Y]/(f, g) < \infty$. $\square$

Since $\mathbb{C}[X, Y]/(f, g)$ is finite dimensional and hence Artinian, it is the direct sum of its localizations at maximal ideals, by Theorem 8.7 of Atiyah and MacDonald [4, Corollary 10.20]. Thus $\mathbb{C}[X, Y]/(f, g) \cong \bigoplus_{P \in \mathbb{C}^2} \mathbb{C}[X, Y]_{(P)}/(f, g)$, where $\mathbb{C}[X, Y]_{(P)}$ is the ring of rational functions $r/s \in \mathbb{C}(X, Y)$ with $s(P) \neq 0$, and $\mathbb{C}[X, Y]_{(P)}/(f, g) \neq 0$ if and only if $P \in V(f) \cap V(g)$.

If f and g have total degree m and n (respectively) then Bezout’s theorem gives mn as the intersection number of the projective completions of $V(f)$ and $V(g)$ in the projective plane $\mathbb{CP}^2$, with multiplicities for intersections at singularities and points where the curves are tangent. (The multiplicity of the intersection at P is $\dim_{\mathbb{C}} \mathbb{C}[X, Y]_{(P)}/(f, g)$. See [17, p. 112].) The intersection points are all in the affine plane $\mathbb{C}^2$ if and only if the homogeneous parts of highest degree f_m and g_n have no common factors.

A plane curve is *smooth* or *nonsingular* if it has a well-defined tangent at every point, i.e., if and only if the partial derivatives f_X and f_Y do not vanish simultaneously at any point of $V(f)$. Let $\text{Sing}(V(f)) = \{(x, y) \in V(f) | f_X(x, y) = f_Y(x, y) = 0\}$ be the set of singular points of $V(f)$. Then $V(f)$ is smooth if $\text{Sing}(V(f))$ is empty.

Corollary 1. *If f is a square-free polynomial then $\text{Sing}(V(f))$ is finite.*

Proof. If f is square-free then f has no common factors with either of its partial derivatives, and so $\text{Sing}(V(f)) = V(f) \cap V(f_X) \cap V(f_Y)$ is finite. $\square$

The next theorem shows that smoothness is an intrinsic property of the ring $O_{V(f)}$, and does not depend on the embedding of $V(f)$ in $\mathbb{C}^2$. We shall use the following special case of Nakayama's lemma in the proof.

Lemma 3. *Let R be a local ring with maximal ideal $M = (r, s)$. If I is an ideal in R such that $M = I + M^2$ then $I = M$.*

Proof. Since $M = M^2 + I$ we may find $m, n, p, q \in M$ and $i, j \in I$ such that $r = i + mr + ns$ and $s = j + pr + qs$. Since the determinant $(1 - m)(1 - q) - pn$ is not in M it is invertible in R , and so we may solve this pair of equations for r and s in terms of i and j . Hence $M \leq I$ and so $M = I$. $\square$

Theorem 4. *Let $f \in \mathbb{C}[X, Y]$ be irreducible. Then the following are equivalent:*

- (1) $O_{V(f)} = \mathbb{C}[X, Y]/(f)$ is integrally closed;
- (2) $f \notin M^2$, for all maximal ideals $M < \mathbb{C}[X, Y]$ which contain (f, f_Y) ;
- (3) $(f, f_X, f_Y) = \mathbb{C}[X, Y]$;
- (4) $V(f)$ is nonsingular.

Proof. The nonzero prime ideals of $O_{V(f)}$ correspond bijectively to the maximal ideals of $\mathbb{C}[X, Y]$ which contain f , via the canonical epimorphism from $\mathbb{C}[X, Y]$ to $O_{V(f)}$, and so $O_{V(f)}$ is a 1-dimensional noetherian domain. Therefore $O_{V(f)}$ is integrally closed if and only if each of its localizations is a discrete valuation ring. Let $A = \mathbb{C}[X]$, and let N be a maximal ideal of $O_{V(f)}$, with preimage M in $A[Y] = \mathbb{C}[X, Y]$. (Then $f \in M$.) Let $Q = M \cap A$, $B = A_Q$ and $C = A[Y]_M$. Then B is a discrete valuation ring with maximal ideal QB and C is a local ring with maximal ideal MC generated by Q and g , for some g representing an irreducible factor of the image of f in $(A/Q)[Y] = (B/QB)[Y]$. Since $0 < (f)C < MC$ is a chain of distinct prime ideals, MC cannot be principal. Therefore $MC/(MC)^2$ has dimension 2 as a vector space over $C/MC \cong \mathbb{C}$, by Nakayama's lemma. The maximal ideal of the localization of $O_{V(f)}$ at N is $MC/(f)$ and so is principal if and only if there is some $t \in C$ such that $MC = (f, t)$. In this case the images of f and t in $MC/(MC)^2$ form a basis, so $f \notin M^2$. Thus (1) $\Rightarrow$ (2).

If $(f, f_X, f_Y) \leq M = (X - \alpha, Y - \beta)$ then on considering the Taylor expansion of f around $(\alpha, \beta) \in \mathbb{C}^2$, we see that $f \in M^2$. Hence (2) $\Rightarrow$ (3). Conversely, if $f \in M^2$ then f_X and f_Y are in M , by an easy application of the Leibniz formula for derivatives of products, so (3) $\Rightarrow$ (2). Moreover, if (2) holds and M is any maximal ideal containing f then $f \notin M^2$. Hence $MC/((f) + (MC)^2)$ is 1-dimensional and so $MC/(f)$ is principal, by Nakayama's lemma again. Therefore (2) $\Rightarrow$ (1).

The equivalence of (3) and (4) is an immediate consequence of the Nullstellensatz. $\square$

Corollary 2. $V(f)$ is smooth if and only if $O_{V(f)}$ is integrally closed.

Let R be an integral domain and let $f, g \in R[Y]$. The resultant of f and g is the product $\text{Res}(f, g) = (f_m)^n (g_n)^m \prod (\alpha_i - \beta_j) = \prod_{i=1}^{i=m} g(\alpha_i)$, where $f = f_m \prod_{i=1}^{i=m} (Y - \alpha_i)$ and $g = g_n \prod_{j=1}^{j=n} (Y - \beta_j)$ in some field containing R . Then $\text{Res}(f, g)$ is in R , depends only on the residue class of g modulo f and is 0 if and only if f and g have a common factor. Moreover $\text{Res}(f, g) = (-1)^{mn} \text{Res}(g, f)$, and formation of resultants is compatible with change of coefficient domains. The discriminant of f is $\text{disc}(f) = (-1)^{\binom{n}{2}} \text{Res}(f, f_Y)$.

Lemma 5. Let R be an integral domain and let $f \in R[Y]$ be a monic polynomial of degree m . If $g \in R[Y]$ then $\text{Res}(f, g)$ is the determinant of the endomorphism of $R[Y]/(f) \cong R^m$ given by multiplication by g .

Proof. We may extend coefficients to a field containing R in which f splits into linear factors, and the result is then clear. $\square$

The equivalence (1) $\equiv$ (2) in Theorem 4 goes through with little change if $\mathbb{C}[X]$ is replaced by any integrally closed 1-dimensional noetherian domain R . If M is a maximal ideal in $R[Y]$ and $(f, f_Y) \leq M$ then $\wp = M \cap R$ divides $\text{disc}(f)$ and $M = (\wp, g)$, where g represents an irreducible factor of the image of f in $R/\wp[Y]$. In particular, let $\Phi_n \in \mathbb{Z}[Y]$ be the n th cyclotomic polynomial, and ζ_n a primitive n th root of unity. Since $X^n - 1$ (and hence Φ_n) has distinct roots over any field of characteristic prime to n , the only primes dividing $\text{disc}(\Phi_n)$ are divisors of n . If $n = mq$ with $q = p^r$ and $(m, p) = 1$ then $\Phi_n(X) = \Phi_m(X^q)/\Phi_m(X^{q/p})$ so $\Phi_n \equiv \Phi_m^{p(q)} \pmod{p}$. Moreover $\Phi_n(X)$ divides $\Phi_p(X^{n/p})$ and so $\Phi_n(\zeta_m)$ divides $\Phi_p(1) = p$. Therefore, Φ_n is not in $(p, g)^2$ for any g representing an irreducible factor of Φ_m in $\mathbb{F}_p[Y]$, and so $\mathbb{Z}[\zeta_n] = \mathbb{Z}[Y]/(\Phi_n)$ is the full ring of integers in $\mathbb{Q}(\zeta_n)$. (This argument avoids first computing the discriminant $\text{disc}(\Phi_n)$.)

The equivalence (2) $\equiv$ (3) holds if $\mathbb{C}$ is replaced by any perfect field. (The latter hypothesis is necessary. Let K be a field of characteristic $p > 0$ with an element b which is not a p th power. Then $f(X, Y) = X^p - b$ is irreducible in $K[X]$, so $K[X]/(f)$ is a field and $K[X, Y]/(f)$ is a PID. However (3) fails, as $f_X \equiv f_Y \equiv 0$ and $V(f)$ is everywhere singular.)

In higher dimensions, Zariski showed that an algebraic variety over a perfect field is nonsingular if and only if its local rings are regular. (The notions “regular” and “integrally closed” are equivalent in the 1-dimensional case.) See [16] for a streamlined account of Zariski’s work.

3. Power series

The localization of the polynomial ring $\mathbb{C}[X, Y]$ at O is too small to reflect the topology adequately. For instance, $Y^2 - X^2(X + 1)$ is irreducible in $\mathbb{C}[X, Y]_{(X, Y)}$ but factors as $(Y + X(1 + X)^{\frac{1}{2}})(Y - X(1 + X)^{\frac{1}{2}})$ in $\mathbb{C}\{X, Y\}$, the ring of germs of holomorphic functions at the origin $O \in \mathbb{C}^2$. (This is the ring of power series which converge on some neighbourhood of O .)

From the algebraic point of view it is natural to pass to the completion of this ring with respect to powers of its maximal ideal, which is the formal power series ring $\mathbb{C}[[X, Y]]$. The ring $\mathbb{C}[[X_1, \dots, X_n]]$ of formal power series in n variables is a local domain, with maximal ideal $M = (X_1, \dots, X_n)$, residue field $\mathbb{C}$ and field of fractions $\mathbb{C}((X_1, \dots, X_n))$. It is complete and Hausdorff with respect to the M -adic topology. In particular, $\mathbb{C}[[X]]$ is a complete discrete valuation ring, with valuation $v(f) = \max\{n \mid f \in (X)^n\} = \dim_{\mathbb{C}} \mathbb{C}[[X]]/(f)$, for $f \neq 0$ (see [4, Chapter 10]).

Theorem 6. *Let I be an ideal in $\mathbb{C}[[X, Y]]$. Then the following are equivalent:*

- (1) $\dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/I < \infty$;
- (2) I contains a power of $M = (X, Y)$;
- (3) I is not contained in any proper principal ideal.

Proof. If $\dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/I < \infty$ then $\mathbb{C}[[X, Y]]/I$ has a finite composition series whose subquotients are isomorphic to $\mathbb{C}[[X, Y]]/M \cong \mathbb{C}$, and so $\mathbb{C}[[X, Y]]/I$ is annihilated by a power of M . Hence (1) $\Rightarrow$ (2).

If $M^n \subseteq I$ holds then $\mathbb{C}[[X, Y]]/I$ is generated as a $\mathbb{C}$ -vector space by the images of the monomials of total degree less than n . Moreover, if also $I \subseteq (f)$ then f divides X^n and f divides Y^n , so f is a unit in $\mathbb{C}[[X, Y]]$. Hence (2) $\Rightarrow$ (1) and (3).

If (3) holds then I is not contained in any proper ideal of the localization $\mathbb{C}((X))[[Y]]$, and so we may write 1 as a finite sum $1 = \sum X^{-n_j} r_j$ for some $r_j \in I$ and $n_j \geq 0$. Clearing denominators, we see that $X^n \in I$ for $n \geq n_X$. Similarly, $Y^n \in I$ for $n \geq n_Y$. Hence $M^n \subset I$ for $n \geq n_X + n_Y - 1$, and so (3) $\Rightarrow$ (2). $\square$

If $f \in \mathbb{C}[[X, Y]]$ the *multiplicity* of f is $v(f) = \min\{n \mid f \in M^n\}$. It is easily seen that multiplicity is additive ($v(fg) = v(f) + v(g)$), positive ($v(f) \geq 0$), and is 0 if and only if f is a unit. If $f \neq 0$ its *initial term* is the sum of the terms of total degree $v(f)$, i.e., the lowest nonzero homogeneous part of f .

Corollary 3. *If $\dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/I = d$ then $M^d \subseteq I$.*

Theorem 7. *The ring $\mathbb{C}[[X, Y]]$ is a 2-dimensional noetherian local unique factorization domain.*

Proof. The above properties of multiplicity imply easily that every element of $\mathbb{C}[[X, Y]]$ has a finite factorization into irreducibles. Let f be an irreducible element which divides a product gh in R . If $f = X$ then it is clearly prime, since $\mathbb{C}[[X, Y]]/(X) = \mathbb{C}[[Y]]$ is a domain. Suppose that $f \neq X$. Then f remains irreducible in $\mathbb{C}((X))[[Y]]$, and divides gh there. Since $\mathbb{C}((X))[[Y]]$ is a PID, f must divide one of the factors, g say. On clearing denominators we see that f divides gX^k in $\mathbb{C}[[X, Y]]$, for some $k \geq 0$. It follows easily that f divides g in $\mathbb{C}[[X, Y]]$, and so irreducibles are prime.

Let I be a nonzero ideal in $\mathbb{C}[[X, Y]]$, and let h be the highest common factor of the elements of I . (This is well defined up to units, since $\mathbb{C}[[X, Y]]$ is factorial.) Then $I = (h)J$, where J is contained in no proper principal ideal. Hence J contains M^n , for some $n \geq 0$, by

Theorem 6, and so is generated by the monomials of degree n together with a basis for the finite-dimensional vector space J/M^n . Therefore, I is finitely generated and so $\mathbb{C}[[X, Y]]$ is noetherian.

In particular, if I is prime and $0 \neq I \neq M$ then I is principal. Thus $\mathbb{C}[[X, Y]]$ is 2-dimensional. $\square$

If $f, g \in R[X]$ are polynomials there are unique polynomials q and r with r of degree $< n$ such that $f = gq + r$, by polynomial long division. This may be extended to formal power series over complete local rings, by the Weierstrass division theorem. (The following argument is due to Gersten [18].)

Theorem 8 (WDT). *Let A be a complete local ring with maximal ideal M , such that $\cap_{n \geq 1} M^n = 0$. If $P, D \in A[[Y]]$ and $D \equiv Y^n$ modulo $MA[[Y]]$ there is a unique pair $q \in A[[Y]]$ and $r \in A[Y]$ of degree $< n$ such that $P = Dq + r$.*

Proof. Define a metric d on A by $d(a, b) = 2^{-n}$ if $a - b \in M^n$ but $a - b \notin M^{n+1}$ and extend this to a metric on $A[[Y]]$ by $d(\sum a_i Y^i, \sum b_j Y^j) = \sup_{k \geq 0} d(a_k, b_k)$. Then $A[[Y]]$ is complete with respect to this metric. Define A -linear functions E, F, T from $A[[Y]]$ to itself as follows. If $h \in A[[Y]]$ let $F(h)$ be the polynomial given by the terms of h of degree $< n$ in Y , and write $h = E(h)Y^n + F(h)$. Let $T(h) = h + E(P - Dh)$. Then it is easily seen that $d(T(h), T(k)) \leq \frac{1}{2}d(h, k)$ and so T is a contraction. Since $A[[Y]]$ is a complete metric space T has a unique fixed point q . Then $E(P - Dq) = 0$, so $r = P - Dq$ is a polynomial of degree $< n$. $\square$

The hypothesis $\cap_{n \geq 1} M^n = 0$ is clearly satisfied in $\mathbb{C}[[X, Y]]$, and holds more generally in any noetherian local ring A , by Corollary 10.20 of Atiyah and MacDonald [4, Corollary 10.20]. See also [23, third ed.] for a short self-contained proof due to Manin.

The Weierstrass Preparation Theorem is a direct consequence. If A is a complete local ring a polynomial $g \in A[Y]$ of degree n is *distinguished* if $g = Y^n + g_1 Y^{n-1} + \cdots + g_n$, where the coefficients g_i are in the maximal ideal of A , for $1 \leq i \leq n$.

Theorem 9 (WPT). *Let A be a complete local ring with maximal ideal M , such that $\cap_{n \geq 1} M^n = 0$. If $f \in A[[Y]]$ has nonzero image $\bar{f}$ in $A/M[[Y]]$ then $f = ug$ for some distinguished polynomial $g \in A[Y]$ and unit $u \in A[[Y]]^\times$, which are uniquely determined by f .*

Proof. We may write $\bar{f} = Y^n \bar{v}$, where n is maximal and $\bar{v}$ is a unit in $A/M[[Y]]$. Let $v \in A[[Y]]$ have image $\bar{v}$ in $A/M[[Y]]$. Then v is a unit in $A[[Y]]$. Applying the WDT with $P = Y^n$ and $D = v^{-1}f$ gives q, r such that $Y^n = qv^{-1}f + r$. On reducing modulo M we see that q is a unit and $r \in MA[[Y]]$. Hence $u = q^{-1}v$ is a unit and $g = Y^n - r$ is a distinguished polynomial, such that $f = ug$. The uniqueness follows from the uniqueness in the WDT. $\square$

The above theorems hold also for $\mathbb{C}\{X, Y\}$. In particular, if P, D and f are holomorphic and $D(0, Y) = Y^n$ the power series q, r, g and u given by the WDT and WPT are holomorphic.

These can be deduced from the formal versions by making suitable convergence estimates (cf. [11]) but it is more natural to use the Cauchy integral formula.

Samuel has shown that square-free formal power series are equivalent to polynomials, up to a formal change of coordinates.

Theorem 10 (Samuel [32]). *Let $f \in M = (X, Y) < \mathbb{C}[[X, Y]]$ be square-free. Then there is an automorphism α of $\mathbb{C}[[X, Y]]$ such that $\alpha(f)$ is a polynomial.*

Proof. If $f \notin M^2$ this is an immediate consequence of the Inverse Function Theorem for formal power series. For if $f_X(0) \neq 0$, say, then $\beta(X) = f$, $\beta(Y) = Y$ defines an automorphism of $\mathbb{C}[[X, Y]]$, and $\beta^{-1}(f) = X$.

Suppose that $f \in M^2$, and let $J = (f_X, f_Y)$. Then J contains some power of M , by Theorem 17 below, and so $M^{2k+1} \leq MJ^2$ for some $k \geq 1$. Let P be a polynomial in the coset $f + MJ^2$. (For instance, we may truncate F after terms of degree at most $2k$.) Then $f = P - A$ where $A = a_{11}f_X^2 + (a_{12} + a_{21})f_Xf_Y + a_{22}f_Y^2$, for some $a_{ij} \in M$. We shall show that there is an automorphism α of the form $\alpha(X) = X + u_{11}f_X + u_{12}f_Y$, $\alpha(Y) = Y + u_{21}f_X + u_{22}f_Y$ with $u_{ij} \in M$ and such that $\alpha(f) = P$. We must solve the equation $f(x + u\nabla f) - f(x) = A$. The left-hand side has the form $u_{11}(1 + G_{11})f_X^2 + u_{12}(1 + G_{12})f_Xf_Y + u_{21}(1 + G_{21})f_Yf_X + u_{22}(1 + G_{22})f_Y^2$, where the G_{ij} are power series in the u_{ij} with coefficients in M and with constant term 0. Let $\theta_{ij}(u) = a_{ij}(1 + G_{ij})^{-1}$. Then θ defines a contraction mapping on the set of 2×2 matrices $\mathbb{M}_2(\mathbb{C}[[X, Y]])$, and its unique fixed point gives a solution. $\square$

A similar argument applies in $\mathbb{C}[[X_1, \dots, X_n]]$. If f and g are holomorphic series and $g = \alpha(f)$ for some formal automorphism α then they are equivalent under a holomorphic change of coordinates α [3].

4. Puiseux series

If f is a square-free holomorphic function such that $f(0) = 0$ the pair $(S_\varepsilon^3, S_\varepsilon^3 \cap f^{-1}(0))$ cut out by the sphere of radius ε centred at 0 determines a link $L(f)$ whose ambient isotopy type is independent of ε , for ε small [27]. The topology of such *algebraic links* is well understood, as $L(f)$ is an “iterated torus link”, with one component for each irreducible factor of f in $\mathbb{C}\{X, Y\}$. The link $L(f)$ is determined up to isotopy (among all such links) by the extended Puiseux data for the factors of f , which provide a template for constructing the link iteratively by forming satellites. (This corresponds to the canonical decomposition of the link exterior along essential tori.)

There is no loss of generality in assuming that f is a polynomial, by the results of Samuel ([32] — see Theorem 10 above) and Artin [3]. In particular, if f is irreducible as a polynomial in Y we may solve the equation $f(X, Y) = 0$ for Y in the algebraic closure of $\mathbb{C}((X))$. We shall show first that this algebraic closure may be constructed by adjoining n th roots of the variable X , for all $n > 1$. Thus the roots of this equation are power series in fractional powers of X . The exponents of the nonzero terms determine the Puiseux invariants for f , and we outline how they may be interpreted in terms of Galois theory.

Let $f \in M = (X, Y) < \mathbb{C}[[X, Y]]$ be square-free. Then $A = \mathbb{C}[[X, Y]]/(f)$ is a 1-dimensional noetherian local ring without nilpotent elements. The uniform topology determined by powers of the maximal ideal is Hausdorff (hence metrizable), by Corollary 10.20 of Atiyah and MacDonald [4, Corollary 10.20], and A is complete with respect to this uniform structure. After a linear change of coordinates, if necessary, we may assume that f is a distinguished polynomial of degree $n > 0$ in Y , by the WPT.

Theorem 11. *Let $f \in \mathbb{C}[[X, Y]]$ be an irreducible distinguished polynomial of degree $n > 0$ in Y , and let $\bar{A}$ be the integral closure of $A = \mathbb{C}[[X, Y]]/(f)$ in its field of fractions $K = \mathbb{C}((X))[Y]/(f)$. Then*

- (1) $\bar{A} = \mathbb{C}[[t]]$ and $K = \mathbb{C}((t))$ for some $t \in K$;
- (2) $\bar{A}$ is a free $\mathbb{C}[[X]]$ -module of rank n ;
- (3) $(X)\mathbb{C}[[t]] = (t^n)$;
- (4) $\dim_{\mathbb{C}} \bar{A}/A < \infty$.

Proof. (1)–(3) follow from Proposition II.3 of Serre [36, Proposition 11.3]. Since A is also clearly free of rank n over $\mathbb{C}[[X]]$ the quotient $\bar{A}/A$ is a finitely generated torsion module and so is finite dimensional over $\mathbb{C}$. $\square$

We shall consider the invariant $\delta_A = \dim_{\mathbb{C}} \bar{A}/A$ in more detail in Section 6 below.

Since the prime ideal (X) becomes an n th power in $\mathbb{C}[[t]]$ the extension is totally ramified. In fact we may assume that $t^n = X$, by the following theorem. Let $\hat{\mathbb{Z}} = \varprojlim Z/mnZ \rightarrow Z/nZ$ be the profinite completion of Z , and let $\overline{\mathbb{C}((X))} = \cup_{n \geq 1} \mathbb{C}((X^{\frac{1}{n}}))$.

Theorem 12. *The algebraic closure of $\mathbb{C}((X))$ is $\overline{\mathbb{C}((X))}$, with Galois group $\hat{\mathbb{Z}}$ acting through multiplication of $X^{\frac{1}{n}}$ by n th roots of unity.*

Proof. Let K be a finite extension of $\mathbb{C}((X))$, of degree n . By Theorem 11 the integral closure of $\mathbb{C}[[X]]$ in K is $R = \mathbb{C}[[t]]$, and $(X) = (t)^n$ in R . Therefore $X = t^n u$, where u is a unit. Since u is a power series with nonzero constant term it has an n th root in R , which is also a unit. Then $t_1 = u^{1/n} t$ is another uniformizer for R , with $X = t_1^n$. Hence $K/\mathbb{C}((X))$ is a Galois extension, with cyclic Galois group, acting on t_1 via multiplication by roots of unity. The theorem follows easily. $\square$

By the theorem, if $f \in \mathbb{C}[[X]][Y]$ is an irreducible monic polynomial of degree n in Y and ζ is a primitive n th root of unity then $f = \prod_{i=1}^n (Y - h(\zeta^i X))$, for some $h \in \mathbb{C}[[X^{\frac{1}{n}}]]$. The fractional power series h is well defined up to the Galois action, and shall be called the *Puiseux series* for f . (In the holomorphic context, the algebraic closure of the field of germs of meromorphic functions $\mathbb{C}\{X\}$ is the union $\cup_{n \geq 1} \mathbb{C}\{X^{\frac{1}{n}}\}$. See Exercise 2.8 of Chapter IV of Serre [36, Proposition 11.3].)

Lemma 13. *Let $f \in \mathbb{C}[[X, Y]]$ be nonzero, and let f_v be its initial term. If $f_v = g_m h_n$ where g_m and h_n have no common factor then $f = gh$ for some g and $h \in \mathbb{C}[[X, Y]]$ with initial terms g_m and h_n , respectively.*

Proof. We construct the homogeneous parts of $g = g_m + g_{m+1} + \dots$ and $h = h_n + h_{n+1} + \dots$ recursively. It suffices to show that given k homogeneous of degree $m + n + p$ for some $p > 0$ we can solve $g_m h_{n+p} + g_{m+p} h_n = k$ for g_{m+p} and h_{n+p} . On making the substitution $Y = XZ$ and dividing by X^{m+n+p} this reduces to an inhomogeneous equation in 1 variable, which may be solved by the Euclidean algorithm. (Note that there are many possible solutions.) $\square$

Let x and y denote henceforth the images of X and Y in A .

Theorem 14. *Let f be irreducible. Then $\bar{A} \cong \mathbb{C}[[t]]$ and $(x, y)\bar{A} = t^{v(f)}\bar{A}$.*

Proof. The initial term of f is a product of linear terms, since it is a homogeneous polynomial in $\mathbb{C}[X, Y]$. Since f is irreducible the linear factors must all be equal, by Lemma 13, and so the initial term has the form $(dX + eY)^n$, where $n = v(f)$ and d and e are not both 0. After a linear change of coordinates we may assume that the initial term is Y^n . On applying the WPT we may then assume that $f = Y^n + \sum a_i X^i Y^{n-i}$, where $a_i \in (X)\mathbb{C}[[X]]$ for $1 \leq i \leq n$. Since $x = t^n$ in $\bar{A} \cong \mathbb{C}[[t]]$ and $f(y) = 0$ in $\bar{A}$ we see easily that y must be in $(t)^{n+1}$. Hence $(x, y)\bar{A} = t^n \bar{A} = t^{v(f)} \bar{A}$. $\square$

In particular, if f is a distinguished polynomial of degree n in Y and $v(f) = n$ then $(x, y)\bar{A} = x\bar{A}$.

Let $h(X^{1/n}) = \sum h_k X^{k/n}$ be a Puiseux series for f , and let $I = \{k | h_k \neq 0\}$ be the set of indices corresponding to nonzero coefficients in h . Then $\text{hcf } I$ is relatively prime to n , since $[K : \mathbb{C}((X))] = n$. Let $\beta(1) = \min\{k \in I | k \notin n\mathbb{Z}\}$ and write $\beta(1)/n = m_1/n_1$ where $(m_1, n_1) = 1$. If $n_1 = n$ then $(m_1, n_1) = (\beta(1), n)$ is the unique characteristic pair for f . If $n_1 < n$ then n_1 divides n , and there are indices $k \in I$ such that k/n cannot be expressed as a fraction with denominator n_1 . Let $\beta(2)$ be the first such, and write $n_1 \beta(2)/n = m_2/n_2$, where $(m_2, n_2) = 1$. Continuing in this way, after finitely many steps we obtain characteristic pairs $(m_1, n_1), \dots, (m_g, n_g)$ such that $m_{i-1}n_i < m_i$ for $1 < i \leq g$ and $n_1 \dots n_g = n$. (Conversely, if a finite sequence $(m_1, n_1), \dots, (m_g, n_g)$ satisfies $m_{i-1}n_i < m_i$ for $1 < i \leq g$ then it is the sequence of characteristic pairs associated to a series $h \in \mathbb{C}((X^{1/n}))$, where $n = n_1 \dots n_g$. The product of the Galois conjugates of $y - h$ gives a series $f \in \mathbb{C}[[X, Y]]$ corresponding to the Puiseux data. See also [22] and [15, pp. 56–58].)

These pairs may be interpreted in terms of “higher ramification groups”: let $G = \text{Gal}(K/\mathbb{C}((X))) \cong Z/nZ$ and $G_i = \{\sigma \in G | \sigma(y) \equiv y \pmod{(t^i)}\}$. Then $\beta(1) = \max\{m | G_m = G\}$, and if $G_{\beta(i)+1} \neq 1$ then $\beta(i+1) = \max\{m | G_m = G_{\beta(i)+1}\}$. Clearly $\beta(1) < \dots < \beta(g)$ and $G_{\beta(g)+1} = 1$. The denominators n_i are given by $n_i = [G_{\beta(i)} : G_{\beta(i)+1}]$, and so $n_1 \dots n_g = |G| = n$. (It may be helpful to contemplate a simple example, such as $Y = X^{1/3} + X^{1/2} + X^{3/4}$, with $n = 12$ and characteristic pairs $(1, 3), (3, 2), (9, 2)$.)

Suppose now that $f = \prod_{i=1}^r f_i$ is square-free, with r irreducible factors. On applying Lemma 12 to each such factor we see that the homogeneous part of f of lowest degree is a product of powers of linear terms. (These correspond to the tangent lines to the irreducible components of $V(f)$ at 0.) The extended Puiseux data for f consists of the Puiseux data for the factors together with the “linking numbers” $\ell_{ij} = \dim_{\mathbb{C}} \mathbb{C}[[X, Y]](f_i, f_j)$ (which are the topological linking numbers of the corresponding components of $L(f)$).

Let $A(i) = \mathbb{C}[[X, Y]]/(f_i)$ and let K_i be the field of fractions of $A(i)$, for $1 \leq i \leq r$. The total ring of quotients of A is the localization $K = A_S$, where $S = A - \bigcup_{i=1}^r (f_i)$ is the set of nonzero divisors in A , and A embeds in K . Moreover $\mathbb{C}[[X]] - \{0\} \subseteq S$, since f is a distinguished polynomial in Y . It follows easily that $K = \mathbb{C}((X)) \otimes_{\mathbb{C}[[X]]} A \cong \prod K_i$. The integral closure of A in K is $\bar{A} \cong \bigoplus \bar{A}(i)$.

Let $v_i : A \rightarrow \mathbb{Z}_{\geq 0}$ be the composite of projection from A onto $A(i)$ with the restriction of the canonical valuation on $\bar{A}(i) \cong \mathbb{C}[[t]]$. Then $v_i(g) < \infty$ for all i if and only if f and g have no common factors. The *singularity semigroup* $S(f) = \{(v_1(g), \dots, v_r(g)) | g \in A, v_i(g) < \infty \forall i\}$ is a subsemigroup of $\mathbb{Z}_{\geq 0}^r$, which encodes the Puiseux data for f [37].

5. The Milnor number

Milnor showed that if $f \in \mathbb{C}\{X_1, \dots, X_{n+1}\}$ has an isolated singularity at O then the exterior $S^{2n+1} - L(f)$ of the associated algebraic link fibres over S^1 , and the fibre F is a wedge of n -spheres. The number of spheres is given by the codimension $\mu(f)$ in $\mathbb{C}\{X_1, \dots, X_{n+1}\}$ of the ideal generated by the partial derivatives of f [27]. In general this codimension is finite if and only if f has an isolated singularity at O . We shall verify this for plane curve singularities ($n = 1$) and shall compute $\mu(f)$ when f is a square-free weighted homogeneous polynomial in two variables.

If R is a PID and N is a finitely generated torsion R -module let $\ell_R(N)$ denote its length. (Note that $\ell_{\mathbb{C}[[X]]}(N) = \dim_{\mathbb{C}} N$ when $R = \mathbb{C}[[X]]$.)

Lemma 15. *Let R be a PID and $\theta : R^n \rightarrow R^n$ a monomorphism. Then $\ell_R(R^n/\theta(R^n)) = \ell_R(R/\det(\theta)R)$.*

Proof. Since R is a PID we may apply elementary row operations, with composition E say, to reduce the matrix of θ to upper triangular form $E\theta$. The automorphism E of R^n induces an isomorphism $R^n/\theta(R^n) \cong R^n/E\theta(R^n)$, and $\det(E)$ is a unit in R . The result now follows by induction on n . $\square$

If $f, g \in \mathbb{C}[[X, Y]]$ let $I(f, g) = \dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/(f, g)$. Then $I(f, g) < \infty$ if and only if f and g have no common factor. Let $M(f) = I(f, f_Y)$.

Lemma 16. *Let $f \in \mathbb{C}[[X]][Y]$ be monic (as a polynomial in Y). Then*

- (1) *If f and g have no common factor $I(f, g)$ is the highest power of X dividing $\text{Res}(f, g)$;*
- (2) *if $f = \prod_{i=1}^r f_i$ is square-free $M(f) = \sum M(f_i) + 2\sum_{i < j} I(f_i, f_j)$.*

Proof. Let $R = \mathbb{C}[[X]]$. Then $\mathbb{C}[[X]][Y]/(f) \cong R^n$, and so (1) follows from Lemma 15, since $I(f, g) = \dim_{\mathbb{C}} \mathbb{C}[[X]]/(\text{Res}(f, g))$, which is the highest power of X dividing $\text{Res}(f, g)$.

We have $\text{Res}(fg, (fg)_Y) = \text{Res}(fg, f_Y g + f g_Y) = \text{Res}(f, f_Y g) \text{Res}(g, f g_Y)$. Therefore $\text{Res}(fg, (fg)_Y) = \text{Res}(f, g)^2 \text{Res}(f, f_Y) \text{Res}(g, g_Y)$, so $M(fg) = M(f) + M(g) + 2I(f, g)$, and (2) follows by a finite induction. $\square$

In particular, if f is a distinguished polynomial in Y then $M(f)$ is the highest power of X dividing $\text{disc}(f)$.

The power series f is *nonsingular* if f_X and f_Y do not both vanish at O . As in the global case, f is nonsingular if and only if it is irreducible and $f \notin M^2$, i.e., $v(f) = 1$. This in turn is so if and only if A is a discrete valuation ring, and hence $A = \bar{A}$. (See Theorem 4.) In general, let $J(f) = (f_X, f_Y)$ be the ideal in $\mathbb{C}[[X, Y]]$ generated by the partial derivatives of f . The *Milnor number* of f is $\mu(f) = \dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/J(f) = I(f_X, f_Y)$. Then f is nonsingular if and only if $J(f) = (1)$, i.e., $\mu(f) = 0$.

Theorem 17. *Let $f \in M = (X, Y) < \mathbb{C}[[X, Y]]$. Then the following are equivalent:*

- (1) f is square-free;
- (2) $J(f)$ contains a power of M ;
- (3) (f, f_X, f_Y) contains a power of M ;
- (4) $\mu(f) < \infty$.

Proof. If (2) does not hold then $J(f) \leq (p)$ for some principal prime $(p) < \mathbb{C}[[X, Y]]$, by Theorem 6. If $(p) = (X)$ it is easily seen that X^2 divides f . Otherwise, we may assume that p is a distinguished polynomial of degree $n > 0$ in Y , by the WPT. Suppose first that $p = Y - h(X)$ for some $h \in \mathbb{C}[[X]]$ with $h(0) = 0$. Define a retraction $\rho : \mathbb{C}[[X, Y]] \rightarrow \mathbb{C}[[X]]$ with kernel (p) by $\rho(k) = k(X, h(X))$. Then $\rho(f)_X = \rho(f_X + f_Y h_X) = 0$, so $\rho(f) = f(0, 0) = 0$. Hence $f = gp$ for some $g \in \mathbb{C}[[X, Y]]$. Then $f_Y = g_Y p + g p_Y = g_Y p + g$, so $f_Y \in (p)$ implies that $g \in (p)$ and so $f \in (p^2)$. In general, p factors into Galois-conjugate linear terms $p_i = Y - h(\zeta^i X^{\frac{1}{n}})$, where $h \in \mathbb{C}[[X^{\frac{1}{n}}]]$. The factors p_i are distinct primes in $\mathbb{C}[[X^{\frac{1}{n}}]][Y]$, since p is irreducible in R . Applying the earlier argument to each p_i we see that f is divisible by p^2 , and so (1) $\Rightarrow$ (2).

The implications (2) $\Rightarrow$ (3) $\Rightarrow$ (1) and (2) $\equiv$ (4) are clear. $\square$

In particular, if f is square-free then $f^n \in J(f)$ for n sufficiently large. In fact $f^2 \in J(f)$ for any square-free f , by a difficult result of Briançon and Skoda [8,25].

If f is holomorphic the singularity of f at O is isolated if and only if (f, f_X, f_Y) contains a power of the maximal ideal in $\mathbb{C}\{X, Y\}$. See Proposition 1.2 of Looijenga [26, Proposition 1.2].

We may compute the Milnor number easily in the following important special case. A polynomial $f \in \mathbb{C}[X, Y]$ is *weighted homogeneous of type* $(N; a, b)$ if it is equivariant with respect to the actions of the multiplicative group $\mathbb{C}^\times$ on $\mathbb{C}^2$ and $\mathbb{C}$ given by $\lambda(x, y) = (\lambda^a x, \lambda^b y)$ and $\lambda(z) = \lambda^N z$, for all $x, y, z \in \mathbb{C}$ and $\lambda \in \mathbb{C}^\times$, where a, b and N are positive integers and a and b are relatively prime. In other words, $f(\lambda^a X, \lambda^b Y) = \lambda^N f(X, Y)$, for all $\lambda \in \mathbb{C}$. It is easy to see that any formal power series satisfying this equation is a polynomial, and is a sum of terms $c_j X^j Y^k$ where $aj + bk = N$. Differentiating each side of this equation with respect to λ at $\lambda = 1$ gives $Nf = aXf_X + bYf_Y$ and so $f \in J(f)$. Let $Z = X^{-\frac{b}{a}} Y$. Then we may write $f(X, Y) = X^{\frac{N}{a}} F(Z)$, where $F(Z) \in \mathbb{C}[Z]$. On considering the factorization of $F(Z)$ we see that the irreducible factors of f are of the form X, Y or $Y^a - cX^b$ for some $c \in \mathbb{C}$.

Suppose now that f is monic in Y and g is another weighted homogeneous polynomial, of type $(N'; a, b)$, and $\text{Res}(f, g) \neq 0$. The roots of f in $\overline{\mathbb{C}((X))}$ are of the form $\alpha_i = \gamma_i X^{\frac{b}{a}}$, where $\gamma_i \in \mathbb{C}$, and so $g(\alpha_i) = c_i X^{\frac{N'}{a}}$, for some $c_i \in \mathbb{C} - \{0\}$. Hence $I(f, g) = \frac{NN'}{ab}$. In particular, if f is a square free, weighted homogeneous distinguished polynomial in Y then f_X and f_Y are weighted homogeneous of types $(N-a; a, b)$ and $(N-b; a, b)$, respectively, and so $\mu(f) = \frac{(N-a)(N-b)}{ab}$.

Reiffen showed that a power series f in 2 variables is weighted homogeneous up to change of coordinates if and only if $f \in J(f)$ [29], and this was extended to the many-variable case by Saito [31]. We shall outline a proof of Reiffen's theorem, based on Saito's argument.

Theorem 18. *Let $f \in M = (X, Y) < \mathbb{C}[[X, Y]]$. Then there is a formal change of coordinates $(u, v) = \phi(X, Y)$ such that $f\phi^{-1}$ is weighted homogeneous in the variables u, v if and only if $f \in J(f)$.*

Proof. Since the condition $f \in J(f)$ is invariant under formal change of coordinates it is clearly necessary. Suppose that it holds. Then there are $\alpha, \beta \in R$ such that $f = Df$, where D is the differential operator $\alpha\partial_X + \beta\partial_Y$. On writing $f = \sum_{n \geq 1} f_n$ as the sum of its homogeneous parts and working modulo powers of $M = (X, Y)$ we find that $\alpha = X(A + m_1)$ and $\beta = Y(B + m_2)$ for some $A, B \in \mathbb{C}^\times$ and $m_1, m_2 \in M$.

Suppose first that $m_1 = m_2 = 0$, so $f = AX\partial_X f + BY\partial_Y f$. If the coefficient of $X^k Y^{n-k}$ in f is nonzero we must have $(A - B)k + Bn = 1$. Therefore, either $A = B = \frac{1}{n}$ and $f = f_n$ is homogeneous of total degree n or $A \neq B$ and each homogeneous term f_n is monomial. Suppose that f is not homogeneous, and that $X^j Y^{m-j}$ and $X^k Y^{n-k}$ are monomials with nonzero coefficients in f and with $n \neq m$. Since $Aj + B(m-j) = 1$ and $Ak + B(n-k) = 1$ and $n \neq m$ the vectors $(j, m-j)$ and $(k, n-k)$ cannot be proportional, and so the determinant $jn - km$ is nonzero. Hence this pair of equations determines A and B uniquely, as rational numbers $A = \frac{a}{N}$ and $B = \frac{b}{N}$, say. (We assume that these fractions are written in lowest terms. In particular, $(a, b) = 1$ if $AB \neq 0$.) If $A > 0$ and $B > 0$ then f is weighted homogeneous of type $(N; a, b)$. If $A = 0$ then $B = \frac{1}{N}$ and $f = Y^N g(X)$ for some 1-variable power series g . Let $g = X^p g_1$ where $g_1(0) \neq 1$ and let $(u, v) = \phi(X, Y) = (X, Y(g_1)^{\frac{1}{N}})$. Then $f\phi^{-1} = u^p v^N$. Similarly if $B = 0$. If $AB < 0$ then $f = X^p Y^q h(X^{|b|} Y^{|a|})$ for some 1-variable power series h with $h(0) \neq 0$, and f becomes weighted homogeneous after a simple change of variables.

In general, we may reduce to the above case if we can find a change of variables $\phi(X, Y) = (u, v)$ such that $u \equiv X \pmod{M^2}$, $v \equiv Y \pmod{M^2}$ and $D = Au\partial_u + Bv\partial_v$. The latter condition gives equations $\alpha\partial_X u + \beta\partial_Y u = Au$ and $\alpha\partial_X v + \beta\partial_Y v = Bv$. On writing $u = X + \sum_{m \geq 2} u_m$ and $v = Y + \sum_{n \geq 2} v_n$, where the u_m and v_n are homogeneous of total degree m and n , respectively, these lead to linear recursion equations for the coefficients of the terms u_m and v_n which are always solvable, and have unique solutions if $A > 0$ and $B > 0$. $\square$

The *Tjurina number* $\tau(f) = \dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/(f, f_X, f_Y)$ is a closely related invariant. Clearly $\tau(f) \leq \mu(f)$, with equality if and only if $f \in J(f)$, and it follows easily from Theorem 17 that $\tau(f) < \infty$ if and only if $\mu(f) < \infty$. The polynomial $g = Y^5 + X^2 Y^2 + X^5$

is a simple example of a formal power series which cannot be so obtained, for $\tau(g) = 10 \neq \mu(g) = 11$. (See [13, pp. 94–97].)

6. The conductor

A distinguished polynomial $f \in \mathbb{C}[[X]][Y]$ is nonsingular if and only if $A = \bar{A}$, and so $\delta_A = \dim_{\mathbb{C}}(\bar{A}/A)$ is a numerical measure of the singularity of f at O . This invariant provides the local correction terms for the Plücker/Riemann–Roch formula $g = \binom{d-1}{2} + \sum_{z \in Z} \delta_z$ relating the genus g and degree d of a projective plane curve $V(f)$ with $\text{Sing}(V(f)) = Z$. (See [35, Chapter IV].)

In this section, we shall show first that δ_A is also the codimension in A of $\mathcal{C} = \{r \in A \mid r\bar{A} \leq A\}$, the *conductor* of $\bar{A}$ into A . (This is the largest ideal of A which is also an ideal of $\bar{A}$.) This was first shown geometrically by Apéry [2] and has been extended in various ways [6]. We shall then show that $\delta_A = (\mu(f) + 1 - r)/2$, a result found by Jung [20] long before the topological significance of $\mu(f)$ had been realized.

Let $S = \mathbb{C}[[X]]$ and let $S_0 = \mathbb{C}((X))$ be the field of fractions of S . Let $f = \sum_{j \leq n} f_j Y^j \in S[Y]$ be a square-free monic polynomial of degree n in Y . Let $A = S[Y]/(f)$ and let y be the image of Y in A . Then $K = S_0 \otimes_S A$ is the total ring of quotients of A and we may define an S_0 -linear homomorphism $\tau : K \rightarrow S_0$ by $\tau(sy^j) = 0$ if $0 \leq j < n-1$ and $\tau(sy^{n-1}) = s$, for all $s \in S_0$.

Lemma 19. *Let $k \in K$. Then $k \in A$ if and only if $\tau(ky^j) \in S$, for all $0 \leq j < n$.*

Proof. The condition is clearly necessary, since $\{y^i \mid 0 \leq i \leq n-1\}$ is a basis for A as an S -module. Assume that it holds, and let $k = \sum_{j < n} s_j y^j$ where $s_j \in S_0$ for $0 \leq j < n$. Then $s_{n-1} = \tau(k)$, $s_{n-2} = \tau(y(k - s_{n-1}y^{n-1}))$ and in general $s_{n-i} = \tau(y^{i-1}(k - \sum_{n-i < j < n} s_j y^j))$, for all $0 < i < n$. A finite induction on i shows that all the coefficients are in S , and so $k \in A$. $\square$

Theorem 20. $\dim_{\mathbb{C}} A/\mathcal{C} = \dim_{\mathbb{C}} \bar{A}/A$.

Proof. Let $D(N) = \text{Hom}_S(N, S_0/S)$, for N any S -module of finite length. Then $D(\mathbb{C}) = D(S/(X)) \cong \mathbb{C}$. Since every S -module of finite length has a finite composition series with subquotients $S/(X) \cong \mathbb{C}$, it follows easily that $\dim_{\mathbb{C}} D(N) = \dim_{\mathbb{C}} N$, for any such N . Define a pairing $\lambda : A \times \bar{A} \rightarrow S_0/S$ by $\lambda(a, \alpha) = [\tau(a\alpha)] \in S_0/S$, for all $a \in A$ and $\alpha \in \bar{A}$. As λ determines monomorphisms from $A/\mathcal{C}$ to $D(\bar{A}/A)$ and from $\bar{A}/A$ to $D(A/\mathcal{C})$, by the Lemma, the result follows easily. $\square$

The function $f = Y^2 - X^3$ gives a simple but nontrivial and instructive example. We have $v(f) = 2$, $\text{disc}(f) = -4X^3$ and $f_Y = 2Y$, and $A = \mathbb{C}[[t^2, t^3]] < \bar{A} = \mathbb{C}[[t]]$. Hence $\delta_A = 1$, and $\mathcal{C} = t^2\bar{A} = (x, y)A$. (Note that $\mathcal{C}$ is not principal as an ideal in A .)

Our exposition of Theorem 20 has its roots in the following observation, which is essentially due to Euler. (See [36, Lemma III.2].)

Lemma 21. Let $f \in \mathbb{C}[[X]][Y]$ be irreducible. Then $\tau(a) = \text{tr}_{K/S_0}(a/f_Y(y))$.

Proof. If f is irreducible then $K \cong \mathbb{C}((t))$, where $t^n = x$, by Theorems 10 and 11, and we may identify tr_{K/S_0} with the $\mathbb{C}((x))$ -linear homomorphism from $\mathbb{C}((t))$ to $\mathbb{C}((x))$ determined by $\text{tr}(1) = n$ and $\text{tr}(t^i) = 0$ for $1 \leq i < n$.

Let $\{y_j | 1 \leq j \leq n\}$ be the roots of $f = f(Y)$ in $\mathbb{C}((t))$. Expanding $\frac{1}{f(Y)}$ by the method of partial fractions gives $\frac{1}{f(Y)} = \sum \frac{1}{f_Y(y_j)(Y-y_j)}$. On expanding each side as power series in $\frac{1}{Y}$ and comparing coefficients, we see that $\text{tr}(\frac{y^i}{f_Y}) = 0$ if $0 \leq i < n-1$ and $\text{tr}(\frac{y^{n-1}}{f_Y}) = 1$. $\square$

We shall show next that δ_A may be expressed in terms of $\mu(f)$ and r (the number of irreducible factors of f).

Lemma 22. Let $A = \mathbb{C}[[X, Y]]/(f)$, where $f \in \mathbb{C}[[X, Y]]$ is irreducible, and let $\theta : A^n \rightarrow A^n$ be a monomorphism. Then $\dim_{\mathbb{C}} A^n/\theta(A^n) = \dim_{\mathbb{C}} \bar{A}^n/\bar{\theta}(\bar{A}^n)$.

Proof. The monomorphism θ extends to an endomorphism of $\bar{A}^n$, and gives rise to a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^n & \longrightarrow & \bar{A}^n & \longrightarrow & \bar{A}^n/A^n & \longrightarrow & 0 \\ \downarrow & & \downarrow \theta & & \downarrow \bar{\theta} & & \downarrow [\theta] & & \downarrow \\ 0 & \longrightarrow & A^n & \longrightarrow & \bar{A}^n & \longrightarrow & \bar{A}^n/A^n & \longrightarrow & 0 \end{array}.$$

The “Snake Lemma” [4, Proposition 2.10]) gives an exact sequence

$$0 \rightarrow \text{Ker}([\theta]) \rightarrow A^n/\theta(A^n) \rightarrow \bar{A}^n/\bar{\theta}(\bar{A}^n) \rightarrow \text{Cok}([\theta]) \rightarrow 0,$$

in which the extreme terms have the same dimension, since $\bar{A}^n/A^n = (\bar{A}/A)^n$ is finite dimensional, by Theorem 10. The result follows easily. $\square$

Lemma 23. Let $f = \prod_{i=1}^r f_i$ be square-free, with r irreducible factors. Then

- (1) $\delta_A = \sum \delta_{A(i)} + \sum_{i < j} I(f_i, f_j)$, where $A(i) = \mathbb{C}[[X, Y]]/(f_i)$ for $1 \leq i \leq r$;
- (2) if $f \in \mathbb{C}[[X]][Y]$ is irreducible and monic in Y then $M(f) = 2\delta_A + n - 1$.

Proof. The natural homomorphism from A to $\oplus A(i)$ is injective and its cokernel is $\oplus_{i < j} (\mathbb{C}[[X, Y]]/(f_i, f_j))$. Since $\bar{A} = \oplus \bar{A}(i)$ (1) follows immediately.

Assume now that f is in $\mathbb{C}[[X]][Y]$ and is irreducible. Then $\bar{A}$ is a domain. Let $\text{tr} = \text{tr}_{K/S_0}$. If P is an A -submodule of $\mathbb{C}((t))$ let $P^* = \{k \in \mathbb{C}((t)) | \text{tr}(kp) \in A \ \forall p \in P\}$. Now $\text{tr}(t^{-n}) = \frac{n}{x}$ and $\text{tr}(t^i) \in \mathbb{C}[[X]]$ if $i > -n$, so $\bar{A}^* = t^{1-n}\bar{A}$. The $n \times n$ matrix with (i, j) entry $\text{tr}(\frac{y^{i+j-2}}{f_Y})$ is 0 above the secondary diagonal, and is 1 along that diagonal. Since $\{y^i | 0 \leq i \leq n-1\}$ is a basis for A as a $\mathbb{C}[[X]]$ -module it follows that $A^* = f_Y^{-1}A$.

Hence $c \in \mathcal{C} \Leftrightarrow c\bar{A} \leq A \Leftrightarrow cf_Y^{-1}\bar{A} \leq A^* \Leftrightarrow \text{tr}(cf_Y^{-1}\alpha a) \in A$ for all $\alpha \in \bar{A}$ and $a \in A \Leftrightarrow cf_Y^{-1} \in \bar{A}^* \Leftrightarrow c \in f_Y t^{1-n}\bar{A}$. Since $\mathcal{C} = t^{2\delta_A}\bar{A}$, by Theorem 20, and $M(f) = \dim_{\mathbb{C}} \bar{A}/f_Y \bar{A}$ (the t -adic valuation of f_Y), by Lemma 22, it follows that $M(f) = 2\delta_A + n - 1$. $\square$

The computation of the conductor is based on Proposition III.11 of Serre [36]. The next lemma is from Risler [30].

Lemma 24. *If $f \in \mathbb{C}[[X]][Y]$ is monic (as a polynomial in Y) then $I(f, f_Y) = I(mf + Xf_X, f_Y)$ for all $m \in \mathbb{Z}$.*

Proof. It shall suffice to prove that $I(f, g) = I(mf + Xf_X, g)$ for each irreducible factor g of f_Y . The integral closure of $\mathbb{C}[[X]][Y]/(g)$ in its field of fractions is isomorphic to $\mathbb{C}[[u]]$, by Theorem 10; let v be the associated valuation. Then $I(f, g) = v(f)$ and $I(mf + Xf_X, g) = v(mf + Xf_X)$, as in Lemma 22. Now $mf + Xf_X = X^{1-m}d(X^m f)/dX$ and so $v(mf + Xf_X) = (1-m)v(X) + v(d(X^m f)/dX)$. As $dw/du = dw/dX \cdot dX/du$ it follows that $v(dw/dX) = v(dw/du) - v(dX/du) = v(w) - 1 - (v(X) - 1) = v(w) - v(X)$. Therefore $v(d(X^m f)/dX) = (m-1)v(X) + v(f)$ and so $I(mf + Xf_X, g) = v(f) = I(f, g)$. $\square$

Suppose that $f = Y^n + \sum a_i X^i Y^{n-i}$ is a distinguished polynomial of degree $n = v(f)$ in Y . Let $Y = XZ$ and define f^σ by $f(X, XZ) = X^n f^\sigma(X, Z)$. Then $f^\sigma = Z^n + \sum a_i Z^{n-i}$ is the *strict (quadratic) transform* of f . Moreover if f is irreducible then so is f^σ .

Quadratic transform is semi-local, in that it replaces the origin by a projective line, and separates the lines through the origin. If g is irreducible and has initial term $(Y - \lambda X)^m$ (i.e., has tangent line $Y - \lambda X = 0$) then $g^\sigma \equiv (Z - \lambda)^m \pmod{X}$ and g^σ determines the germ of a curve through $(X, Z) = (0, \lambda)$. It follows that if $f = \prod_{i=1}^r f_i$ and $g = \prod_{j=1}^s g_j$ then $I(f, g) = v(f)v(g) + \sum \sum I(f_i^\sigma, g_j^\sigma)$, (with only factors having common tangents contributing to the double sum). Let $\tilde{I}(f^\sigma, g^\sigma)$ denote the latter sum.

Theorem 25. *Let $f = \prod_{i=1}^r f_i$ be square-free, with r irreducible factors. Then $\delta_A = (\mu(f) + 1 - r)/2$.*

Proof. We may assume that $f = Y^n + \sum a_i X^i Y^{n-i}$ is a distinguished polynomial of degree $n = v(f) > 1$ in Y , and (hence) that each factor f_i is also a distinguished polynomial of degree n_i in Y , for $1 \leq i \leq r$. Differentiating the equation $X^n f^\sigma(X, Z) = f(X, XZ)$ with respect to Z gives $X^n (f^\sigma)_Z = Xf_Y$ (so $(f_Y)^\sigma = (f^\sigma)_Z = f_Z^\sigma$, say) and so $M(f) = I(f, f_Y) = n(n-1) + \tilde{I}(f^\sigma, f_Z^\sigma)$. Differentiation with respect to X gives $nX^{n-1}f^\sigma + X^n(f^\sigma)_X = f_X + Zf_Y$, and so $\mu(f) = I(f_X, f_Y) = I(f_X + Yf_Y, f_Y) = (n-1)^2 + \tilde{I}(nf^\sigma + X(f^\sigma)_X + (X-1)Zf_Z^\sigma, f_Z^\sigma)$. (Note that since f is square-free $f_X \neq 0$ and $v(f_X) = n-1$.) This in turn equals $(n-1)^2 + \tilde{I}(f^\sigma, f_Z^\sigma)$, by Lemma 24. Hence $\mu(f) = M(f) + 1 - n$. Now $M(f) = \sum M(f_i) + 2\sum_{i < j} I(f_i, f_j) = 2(\sum \delta_{A(i)} + \sum_{i < j} I(f_i, f_j)) + (\sum n_i) - r = 2\delta_A + n - r$, by Lemmas 16 and 22. Therefore $\mu(f) = 2\delta_A - r + 1$ and so $\delta_A = (\mu(f) + 1 - r)/2$. $\square$

This argument is taken from Risler [30]; the reference to Jung [20] (pp. 368–370) is from MR46#5334. Milnor has given a topological argument for Theorem 25 (for f a polynomial), in which the projective completion of $V(f)$ is approximated by other related curves. (See [27, Theorem 10.5].)

7. Resolution of singularities

To resolve the singularities of an algebraic variety V means to give a nonsingular variety V' and a morphism $p : V' \rightarrow V$ which restricts to an isomorphism over the nonsingular points of V . This issue is usually considered in the context of projective varieties.

If $V(f)$ is an irreducible plane curve let $\tilde{O}_{V(f)}$ be the integral closure of the domain $O_{V(f)}$ in its field of fractions. Then $\tilde{O}_{V(f)}$ is finitely generated as an $O_{V(f)}$ -module, and hence as a $\mathbb{C}$ -algebra. Hence it is the coordinate ring of a smooth algebraic curve $\tilde{V}$ in a (possibly) higher-dimensional affine space $\mathbb{C}^m$. Maximal ideals of $\tilde{O}_{V(f)}$ restrict to maximal ideals of $O_{V(f)}$, so there is a canonical map from $\tilde{V}$ to $V(f)$. However, it may not be possible to find such a “smooth model” which is also a plane curve.

For curves the singular set is finite, and the problem may also be considered locally. As observed above, resolution by quadratic transform is semi-local. Thus after a quadratic transform of f centred at $Y = 0$ we may have to apply further quadratic transforms centred at $Z = p_i$, where p_i is the slope of a tangent to $V(f)$ at 0.

The local approach actually gives more information than normalization (passage to the integral closure). Assume that f is a distinguished polynomial of degree $n > 0$ in Y and let $B = \mathbb{C}[[X]][Z]/(f^\sigma) \cong A[y/x] \leq \bar{A}$, where $f^\sigma(X, Z) = X^{-n} f(X, XZ)$. Now $f_Z^\sigma = X^{1-n} f_Y$, by the chain rule, so $M(f^\sigma) = M(f) - n(n-1)$ and $\delta_B = \delta_A - (n(n-1))/2$. Thus after finitely many such quadratic transforms (possibly followed by linear changes of coordinate) we obtain a nonsingular f^ω . This process of resolution by quadratic transformations is canonical, and if f is irreducible the sequence of multiplicities $v(f) \geq v(f^\sigma) \geq \dots \geq v(f^\omega) = 1$ determines f up to topological equivalence. (It is conventional to truncate the sequence before the first “1”.) The analogous invariant in the reducible case is somewhat more complex, as one must keep track of how the branches with common tangents interact. (See [10, p. 502ff].)

Let $\psi_N : X_N \rightarrow X_{N-1} \rightarrow \dots \rightarrow X_0 = \text{Spec}(\mathbb{C}[[X, Y]])$ be the sequence of quadratic transforms corresponding to the canonical resolution of f , let $E_N = \psi_N^{-1}(0)$ and let $V_N = \psi_N^{-1}(V) - E_N$ be the strict preimage of $V(f)$ in X_N . The *resolution graph* Γ_f is the weighted graph with one vertex for each irreducible component of $\psi_N^{-1}(V) = E_N \cup V_N$, an edge joining vertices corresponding to components which meet, and weights $i(e) = 1 + \max\{i | e \text{ has image } 1 \text{ point} \in X_i\}$ on each vertex e corresponding to a component of E_N . (These weights are the negatives of the self-intersection numbers of the components e in X_N .) An algorithm for determining the resolution graph for an irreducible f from its Puiseux data is given in [12]. Conversely, each of the multiplicity sequence, the resolution graph and the extended Puiseux data (i.e., including the linking numbers) determines the other sets of invariants (see [10, p. 512ff]).

8. The relative de Rham module and the Gauß–Manin connection

In principle, topological invariants of an algebraic link $L(f)$ can be computed from the Puiseux data (or, equivalently, from the resolution graph Γ_f), and this has largely been done in [15]. (In particular, if f and g are irreducible and relatively prime then

$\text{lk}(L(f), L(g)) = \dim_{\mathbb{C}} \mathbb{C}[[X, Y]]/(f, g)$, and is strictly positive.) The link $L(f)$ is in fact determined among all such links by its component knots and their linking numbers (see [39, Lemma 7.1]), and hence by its multivariable Alexander polynomial $\Delta_1(L(f))$ [38]. The invariant δ_A is the unknotting number of $L(f)$; this is a consequence of the Thom conjecture (see the survey article [7]), which has been proven by Kronheimer and Mrowka [21].

The (reduced) Alexander polynomial of a fibred link is the characteristic polynomial of the monodromy, acting on the cohomology $H^1(F; \mathbb{C})$ of the fibre F . Using the deep coherence theorem of Grauert, Brieskorn has shown that for an algebraic link $H^1(F; \mathbb{C})$ may be identified with the kernel of the topological Gauß–Manin connection over a deleted neighbourhood of 0 in $\mathbb{C}$, and that the Alexander polynomial may (in principle) be computed in terms of an algebraically defined Gauß–Manin connection on a relative de Rham cohomology module [9]. (His arguments apply also in the many variable cases.) We shall use elementary calculations to recover Brieskorn’s results for the case when f is a weighted homogeneous polynomial in two variables.

Let $R = \mathbb{C}[[X, Y]]$ and let $f = \sum_{k \leq n} a_k(X)Y^k$ be a square-free distinguished polynomial of degree n in Y . Let $f^* : S = \mathbb{C}[[s]] \rightarrow R$ be the ring homomorphism sending s to f , defined by $f^*(g) = g \circ f$. Then $R \cong S[[X, Y]]/(f - s)$ and $A = \mathbb{C}[[X, Y]]/(f) \cong R/(s)$. Let $\Omega^1 = R dX \oplus R dY$ be the module of 1-differentials of R over $\mathbb{C}$ and let $\Omega^0 = R$ and $\Omega^2 = \Omega^1 \wedge \Omega^1 = R dX \wedge dY$ be the other nonzero exterior powers of Ω^1 . (Thus Ω^p is the completion of the module of germs of holomorphic p -forms at O in $\mathbb{C}^2$.) The cochain complex Ω^* determined by the exterior derivatives $d : \Omega^p \rightarrow \Omega^{p+1}$ gives a resolution $0 \rightarrow \mathbb{C} \rightarrow R \rightarrow \Omega^1 \rightarrow \Omega^2 \rightarrow 0$.

Let $\Omega_f^p = \Omega^p / df \wedge \Omega^{p-1}$. The exterior derivative on Ω^p induces S -linear differentials $d_f : \Omega_f^p \rightarrow \Omega_f^{p+1}$ (via f^*) and so we obtain a S -cochain complex Ω_f^* . Let $H_f^1 = H^1(\Omega_f^*)$, $H' = \Omega_f^1 / dR = \Omega^1 / dR + R df$ and $H'' = \Omega^2 / df \wedge dR$. Then H_f^1 , H' and H'' are S -modules, $H_f^1 \leq H'$ and $H'' \cong \text{Cok}(\delta)$, where $\delta : R \rightarrow R$ is the S -linear derivation given by $\delta(g) = g_X f_Y - g_Y f_X$, for all $g \in R$. Since δ is S -linear and $\text{Im}(\delta) \leq J(f)$ it induces a $\mathbb{C}$ -linear derivation $\delta_A : A \rightarrow A$, with $\text{Im}(\delta_A) \leq J_A$, where J_A is the image of $J(f)$ in A .

Since f_X and f_Y are relatively prime wedge product with df induces a monomorphism $\kappa : H' \rightarrow H''$, with cokernel $\Omega_f^2 \cong R/(f_X, f_Y)$. Moreover d induces a $\mathbb{C}$ -linear bijection $d' : H' \rightarrow H''$, with $d'(H_f^1) = \kappa(H')$. (For if $d\eta = df \wedge dg$ then $d(\eta + gdf) = 0$ and so $\eta + gdf = dr$ for some $r \in R$, since Ω^* is exact above degree 0.) This induces an S -linear isomorphism $H'/H_f^1 \cong \text{Cok}(\kappa) \cong \Omega_f^2$.

A meromorphic connection on a $\mathbb{C}[[s]]$ -module N is a $\mathbb{C}$ -linear function $\Delta : N \rightarrow N[s^{-1}] = \mathbb{C}((s)) \otimes_{\mathbb{C}[[s]]} N$ such that $\Delta(gm) = g\Delta(m) + (\frac{d}{ds}g)m$ for all $g \in \mathbb{C}[[s]]$ and $m \in N$. Such a connection Δ is regular if $\dim_{\mathbb{C}((s))} N[s^{-1}] < \infty$ and $s\Delta$ maps a lattice in $N[s^{-1}]$ into itself.

The $\mathbb{C}[[s]]$ -module H_f^1 supports a natural meromorphic connection, defined as follows. If $\eta \in \Omega^1$ represents a class in H_f^1 then $d\eta = df \wedge \psi = \kappa([\psi])$, for some 1-form ψ . The class of ψ in H' is well-defined, and $d(f^k \psi) = f^k d\psi + k f^{k-1} df \wedge \psi$ is in $df \wedge \Omega^1 = (f_X, f_Y) dX \wedge dY$, for k large, since $f^k \in (f_X, f_Y)$ for k large, by Theorem 17. Hence $f^k \psi \in H_f^1$, and the function defined by $\nabla([\eta]) = s^{-k}[f^k \psi]$ gives a well-defined meromorphic connection on H_f^1 . This is the local Gauß–Manin connection ∇ associated to f .

Brieskorn used the coherence theorem of Grauert to show that the S -modules H , H' and H'' are finitely generated and of rank $\mu(f)$, and that the local Gauß–Manin connection of an isolated singularity is always regular [9]. Moreover H'' is torsion free as an S -module [34]. Hence so are H_f^1 and H' . If we identify H_f^1 and H' with submodules of $H_f^1[s^{-1}]$ we have $\nabla([\eta]) = \kappa^{-1}d'([\eta])$, and so ∇ maps H_f^1 bijectively onto H' .

The apparent contradiction of the injectivity of the local Gauß–Manin connection as just defined with the claim that the cohomology of the Milnor fibre may be identified with the kernel of the topological Gauß–Manin connection may be resolved by interpreting the equation $s\nabla(\eta) = 0$ as a linear system of first-order ODEs and extending coefficients from S to a larger ring $\widehat{S}$, to include all possible solutions to such a linear system. Let $\widehat{S} = \overline{\mathbb{C}((s))}[\lambda]$, where $\overline{\mathbb{C}((s))} = S[s^{-1}, s^{\frac{1}{n}}; n > 1][\lambda]$ is the algebraic closure of the field of fractions of S , and extend the derivation $\frac{d}{ds}$ so that $\frac{d}{ds}(s^k) = ks^{k-1}$ for all rational exponents k and $\frac{d}{ds}(\lambda) = s^{-1}$. (Thus λ corresponds to $\log(s)$.) Fix a basis $\{v_1, \dots, v_\mu\}$ for H_f^1 over S , and write $s\nabla(v_i) = \sum_{j \leq \mu} s_{ij} v_j$. Let $e_1, \dots, e_\mu \in \widehat{S}$. Then

$$s\nabla(\sum_{i \leq \mu} e_i v_i) = \sum_{i \leq \mu} \left(\left(s \frac{d}{ds} e_i \right) v_i + e_i \sum_{j \leq \mu} s_{ij} v_j \right),$$

and so the equation $s\nabla(\eta) = 0$ corresponds to the linear system

$$s \frac{d}{ds} e_i + \sum_{k \leq \mu} s_{ki} e_k = 0 \quad (1 \leq i \leq \mu)$$

over $\widehat{S}$. The monodromy of this linear system is essentially the cohomological monodromy (with complex coefficients) of the Milnor fibration [9].

9. The weighted homogeneous case

In this section, we shall establish the results of Brieskorn and Sebastiani for the special case of weighted homogeneous polynomials in two variables.

Theorem 26. *Let f be a square-free distinguished polynomial of degree n in Y which is weighted homogeneous of type $(N; a, b)$. Then $H'' \cong S^{\mu(f)}$, where $\mu(f) = \frac{(N-a)(N-b)}{ab}$.*

Proof. Since $aXf_X + bYf_Y = Nf$ we see that $\delta(Y^j) = -jY^{j-1}f_X$ if $0 < j < n$ and $\delta(X^i Y^j) = \frac{ai+bj}{a} X^{i-1} Y^j f_Y - \frac{jN}{a} X^{i-1} Y^{j-1} f$ if $0 < i < \infty$. On passing to $A = R/(s)$ we get $x^i y^j f_Y = \delta_A(\frac{a}{ai+bj+a} x^{i+1} y^j)$ and $x^i y^j f_X = \delta_A(\frac{b}{ai+bj+b} x^i y^{j+1})$ for all $i, j \geq 0$. Hence $\text{Im}(\delta_A) = J_A$. Since $f \in J(f)$ we have $R/J(f) \cong A/J_A = A/\text{Im}(\delta_A) \cong R/(sR + \text{Im}(\delta))$. Moreover $sR + \text{Im}(\delta) \leq J(f)$ and so $sR + \text{Im}(\delta) = J(f)$. Let $U = \{u_i | 1 \leq i \leq \mu(f)\} \subset R$ represent a basis for $R/J(f)$ as a $\mathbb{C}$ -vector space. We may choose functions $h, k : R \rightarrow R$ so that if $g \in R$ then $g = \delta(h(g)) + sk(g) + \sum c_i(g)u_i$, for some coefficients $c_i(g) \in \mathbb{C}$. Applying a similar expansion to $k(g)$ and iterating, we conclude that $g = \sigma(g) + \delta(\tilde{h}(g))$, where $\sigma(g) = \sum_{m \geq 0} c_i(k^m(g))s^m$ and $\tilde{h}(g) = \sum_{0 \leq m} s^m h(k^m(g))$, and so $R = SU + \text{Im}(\delta)$. Hence $H'' \cong \text{Cok}(\delta)$ is generated by U as an S -module.

The ring R is a free $S[[X]]$ -module with basis $\{1, \dots, Y^{n-1}\}$, and so every element $g \in R$ is uniquely expressible as a sum $g = \sum_{0 \leq i} \sum_{j=0}^{n-1} g_{ij} X^i Y^j$, with coefficients in S . Since $\delta(1) = 0$ it follows that $\delta(g) = \sum' g_{ij} \delta(X^i Y^j)$, where $\sum'$ denotes summation over indices (i, j) with $0 \leq i < \infty$, $0 \leq j < n$ and $0 < i + j$. Thus if H'' is not free of rank $\mu(f)$ as an S -module there is a nontrivial linear relation $\sum_{k=1}^{k=\mu(f)} e_k u_k = \sum' g_{i,j} \delta(X^i Y^j)$ in R , with coefficients $\{e_k\}$ and $\{g_{i,j}\}$ in S and not all divisible by s . Since $\sum_{k=1}^{k=\mu(f)} e_k(0) u_k = 0$ in $R/J(f)$ we see that $e_k(0) = 0$ for all $k \leq \mu(f)$. Hence $\sum' g_{i,j}(0) \delta_A(x^i y^j) = 0$ in A . Now $\delta_A(1) = 0$, $\delta_A(y^j) = -j y^{j-1} f_X$ if $0 < j < n$ and $\delta_A(x^i y^j) = \frac{ai+bj}{a} x^{i-1} y^j f_Y$ if $0 < i < \infty$, while $a x f_X + b y f_Y = 0$. Hence $0 = x(\sum' g_{i,j}(0) \delta_A(x^i y^j)) = \sum_{j=0}^{j=n-1} g_{0,j}(0) x \delta(y^j) + \sum_{1 \leq i} \sum_{j=0}^{j=n-1} g_{i,j}(0) x \delta(x^i y^j) = (\sum' g_{i,j}(0) \frac{ai+bj}{a} x^i y^j) f_Y$. But $\{1, \dots, y^{n-1}\}$ is a basis for A as a free $\mathbb{C}[[x]]$ -module, and f_Y is a nonzero divisor in A . Therefore the coefficients $g_{i,j}(0)$ are all 0 also. This contradicts our assumption and so $H'' \cong S^{\mu(f)}$. That $\mu(f) = \frac{(N-a)(N-b)}{ab}$ was shown in Section 5 above. $\square$

Since $f \in (f_X, f_Y)$ we have $f \Omega^2 \leq df \wedge \Omega^1$, and so H''/sH'' maps onto $\Omega_f^2 \cong R/J(f)$. Since these $\mathbb{C}$ -vector spaces each have dimension $\mu(f)$ this epimorphism must be an isomorphism. Hence $sH'' = df \wedge \Omega^1 = \kappa(H')$. Since $H'/H_f^1 \cong \Omega_f^2$ also it follows that $H_f^1 = sH'$. Moreover $d(f\psi) \in df \wedge \Omega^1$ for all $\psi \in \Omega^1$ and so ∇ is clearly regular.

Suppose that $f = g f_X + h f_Y$ and let $\alpha = -h dX + g dY$ and $\omega = dX \wedge dY$. Then $\kappa(\alpha) = [df \wedge \alpha] = [f\omega]$. Let $U \subset R$ represent an S -basis for H'' . Then $\{u\alpha | u \in U\}$ is a basis for H' and so $\{uf\alpha | u \in U\}$ is a basis for H_f^1 . It is easily verified that $d(uf\alpha) = (u + (gu)_X + (hu)_Y) f\omega$ for all $u \in R$. (Note that we may always assume that U is a set of monomials $\{X^i Y^j\}$.)

If $f = Y^n + X^m$ we may take $\alpha = \frac{1}{n}(-Y dX + X dY)$. Let $\eta_{i,j} = [f X^i Y^j \alpha]$. Then $\{\eta_{i,j} | 0 \leq i < m-1, 0 \leq j < n-1\}$ is a basis for H_f^1 , and we see easily that $\nabla(\eta_{i,j}) = [\lambda(i, j) X^i Y^j \alpha] = s^{-1} \lambda(i, j) \eta_{i,j}$, where $\lambda(i, j) = 1 + \frac{i+1}{m} + \frac{j+1}{n}$. The corresponding system of ODEs $s \frac{d}{ds}(e_{i,j}) + \lambda(i, j) e_{i,j} = 0$ has basic solutions $e_{i,j} = s^{-\lambda(i,j)}$. As s moves once around $0 \in \mathbb{C}$ these solutions change by $\zeta^{-m(j+1)-n(i+1)}$, where $\zeta = \exp(2\pi i/mn)$. In particular, if $(m, n) = 1$ then as i, j vary in the ranges $0 \leq i < m-1$ and $0 \leq j < n-1$ these factors $\zeta^{-m(j+1)-n(i+1)}$ run through the (mn) th roots of unity which are neither m th nor n th roots of unity. Hence the characteristic polynomial of the monodromy is $(t^{mn} - 1)(t - 1)/(t^m - 1)(t^n - 1)$, which is the Alexander polynomial of the (m, n) -torus knot.

If f is weighted homogeneous of type $(N; a, b)$ we may take $g = \frac{a}{N} X$ and $h = \frac{b}{N} Y$, to get $d\alpha = \frac{a+b}{N} \omega$. In particular,

$$d(X^i Y^j f \alpha) = \left(1 + \frac{a(i+1)}{N} + \frac{b(j+1)}{N}\right) X^i Y^j f \omega.$$

The monodromy is again diagonalizable and its eigenvalues are roots of unity, and so it has finite order. However, it is not immediately obvious how to choose monomials representing a $\mathbb{C}$ -basis for $R/J(f)$.

Since finite generation of $\text{Cok}(\delta)$ is invariant under change of coordinates it follows from Reiffen's theorem (Theorem 18) that Theorem 26 and its consequences apply whenever f

is a square-free distinguished polynomial in R such that $f \in J(f)$. Can the argument for Theorem 26 be extended to establish finiteness for all square-free distinguished polynomials? An ad hoc argument handles polynomials of the form $f = Y^n + a_1Y + a_0$ with $n \geq 3$ and a_0 and a_1 nonzero, and with $(n-1)d(0) < nd(1)$, where $d(i)$ is the highest power of X dividing a_i , for $i = 0$ or 1 .

If $f \notin (f_X, f_Y)$ then establishing regularity of ∇ may be more difficult. For if $A \in R$ is such that $A_X = Xf_X$ then $d(AdY) = df \wedge (-XdY)$, so $\omega = AdY$ represents a class in H_f^1 with $\nabla(\omega) = -XdY$, but $d(fXdY) = fdX \wedge dY + Xdf \wedge dY$ is not in $df \wedge \Omega^1$.

In general, the eigenvalues of the monodromy are always roots of unity, and the largest Jordan block size is 2. (See [33] for a derivation of this from the Briançon–Skoda result that $f^2 \in J(f)$.) The monodromy has finite order if f is irreducible [24]. On the other hand the monodromy of $(Y^3 + X^2)(Y^2 + X^3)$ has infinite order and so is not diagonalizable [1].

10. An hermitean pairing

The Seifert form for an algebraic link may also be related to the local Gauß–Manin connection, via integration of C^∞ forms [5]. Barlet defines a finitely generated $\mathbb{C}[[s, \bar{s}]]$ -module $\mathcal{M}$ with an involution extending the involution $s \leftrightarrow \bar{s}$ on the coefficient ring and with a meromorphic connection ∂_s such that $\partial_s(\bar{s}.m) = \bar{s}.\partial_s m$, for all $m \in \mathcal{M}$. He then uses integration of representative compactly supported C^∞ forms to define a pairing $\hat{H}$ on H' with values in $\mathcal{M}$ which is hermitean and horizontal, in the following senses:

- (1) it is S -linear in the first variable;
- (2) $\hat{H}(\alpha, \beta) = \hat{H}(\beta, \alpha)$, for all $\alpha, \beta \in H'$;
- (3) (horizontal) $s^k \partial_s \hat{H}(\alpha, \beta) = \hat{H}(s^k \nabla \alpha, \beta)$, for all $\alpha, \beta \in H'$ and $k \geq 0$.

He shows that $(H', \nabla, \hat{H})$ is essentially equivalent to the intersection form on $H_1(F; \mathbb{C})$ together with the monodromy automorphism $h_{\mathbb{C}}$, after localization away from the 1-eigenspace of the monodromy [5]. The 1-eigenspace is trivial for any knot, and has dimension $r-1$ if f has r irreducible factors [14]. Barlet's argument applies in all dimensions and does not require a resolution of the singularity in its application. On the other hand, it is not clear how to translate it into terms of local commutative algebra.

Acknowledgements

I would like to acknowledge the support of Grey College and the Department of Mathematical Sciences at the University of Durham (through a Grey College Mathematics Fellowship) while revising Section 6 and writing the final three sections of this paper.

References

- [1] N. A'Campo, Sur la monodromie des singularités isolées d'hyper-surfaces complexes, *Invent. Math.* 20 (1973) 147–169.

- [2] R. Apéry, Sur les branches superlinéaires des courbes algébriques, C. R. Acad. Sci. (Paris) 222 (1946) 1198–1200.
- [3] M. Artin, On the solution of analytic equations, Invent. Math. 5 (1968) 277–291.
- [4] M. Atiyah, I.G. MacDonald, Introduction to Commutative Algebra, Addison-Wesley, Reading, Menlo Park, London, Don Mills, 1969.
- [5] D. Barlet, Forme hermitienne canonique sur la cohomologie de la fibre de Milnor d'une hypersurface a singularité isolée, Invent. Math. 81 (1985) 115–153.
- [6] H. Bass, On the ubiquity of Gorenstein rings, Math. Z. 82 (1963) 8–28.
- [7] M. Boileau, C. Weber, Le problème de Milnor sur le nombre gordien des noeuds algébriques, in: C. Weber (Ed.), Noeuds, Tresses et Singularités Monographie, vol. 31, de L'Enseign Mathematics, Geneva, 1983, pp. 49–98.
- [8] J. Briançon, H. Skoda, Sur la clôture intégrale d'un idéal de germes de fonctions holomorphes en un point de $\mathbb{C}^n$, C.R. Acad. Sci. Paris 278 (1974) 949–951.
- [9] E. Brieskorn, Die Monodromie der isolierten Singularitäten von Yperflächen, Manuscripta Math. 2 (1970) 103–161.
- [10] E. Brieskorn, H. Knörrer, Plane Algebraic Curves, Birkhäuser, Verlag, Basel, Boston, Stuttgart, 1986.
- [11] A. Brill, Über den Weierstraßschen Vorbereitungssatz, Math. Ann. 64 (1910) 538–549.
- [12] S.D. Cutkosky, H. Srinivasan, The algebraic fundamental group of a curve singularity, J. Algebra 230 (2000) 101–126.
- [13] A. Dimca, Topics in Real and Complex Singularities, Vieweg Advanced Lectures in Mathematics, Braunschweig, 1987.
- [14] A. Durfee, The characteristic polynomial of the monodromy, Pacific J. Math. 59 (1975) 21–26.
- [15] D. Eisenbud, W.D. Neumann, Three-dimensional Link Theory and Invariants of Plane Curve Singularities, Annals of Mathematics Study, vol. 110, Princeton University Press, Princeton, 1985.
- [16] J.D. Emerson, Simple points on an affine algebraic variety, Amer. Math. Monthly 82 (1975) 132–147.
- [17] W. Fulton, Algebraic Curves. An Introduction to Algebraic Geometry, W.A. Benjamin, New York, Amsterdam, 1969.
- [18] S.M. Gersten, A short proof of the algebraic Weierstrass preparation theorem, Proc. Amer. Math. Soc. 88 (1983) 751–752.
- [19] J.A. Hillman, Polynomials determining Dedekind domains, Bull. Austral. Math. Soc. 29 (1984) 167–175.
- [20] H.W.E. Jung, Einführung in die Theorie der algebraischen Funktionen zwei Veränderlicher, Akademie, Verlag, Berlin, 1951.
- [21] P.B. Kronheimer, T.S. Mrowka, The genus of embedded surfaces in the projective plane, Math. Res. Lett. 1 (1994) 797–808.
- [22] T.-C. Kuo, Generalized Newton–Puiseux theory and Hensel's lemma in $\mathbb{C}[[x, y]]$, Canad. J. Math. 41 (1989) 1101–1116.
- [23] S. Lang, Algebra, revised third ed., Graduate Texts in Mathematics, vol. 211, Springer, Berlin, 2002.
- [24] Lê Dũng Tráng, Sur les noeuds algébriques, Compositio Math. 25 (1972) 281–321.
- [25] J. Lipman, B. Teissier, Pseudorational local rings and a theorem of Briançon–Skoda about integral closures of ideals, Michigan Math. J. 28 (1981) 97–116.
- [26] E. Looijenga, Isolated Singular Points on Complete Intersections, London Mathematical Society Lecture Note Series, vol. 77, Cambridge University Press, Cambridge, 1984.
- [27] J.W. Milnor, Singularities of Complex Hypersurfaces, Annals of Mathematics Study, vol. 61, Princeton University Press, Princeton, NJ, 1968.
- [28] M. Morisita, On certain analogies between knots and primes, J. Reine Angew. Math. 550 (2002) 141–167.
- [29] H.-J. Reiffen, Kontrahierbare eindimensionale Hyperflächen, Nachr. Akad. Wiss. Göttingen II Math. Phys. Kl. 3 (1968) 39–46.
- [30] J.-J. Risler, Sur l'idéal jacobien d'une courbe plane, Bull. Soc. Math. France 99 (1971) 305–311.
- [31] K. Saito, Quasihomogene isolierte Singularitäten von Hyperflächen, Invent. Math. 14 (1971) 123–142.
- [32] P. Samuel, Algébricité de certains points singuliers algébroides, J. Math. Pures Appl. 35 (1956) 1–6.
- [33] J. Scherk, On the monodromy theorem for isolated hypersurface singularities, Invent. Math. 58 (1980) 289–301.
- [34] M. Sebastiani, Preuve d'une conjecture de Brieskorn, Manuscripta Math. 2 (1970) 301–308.
- [35] J.-P. Serre, Groupes Algébriques et Corps de Classe, Hermann et Cie, Paris, 1959.
- [36] J.-P. Serre, Corps Locaux, Hermann et Cie, Paris, 1962.
- [37] R. Waldi, Wertehalbgruppe und Singularität einer ebene algebraischen Kurve, Dissertation, Regensburg, 1972.
- [38] M. Yamamoto, Classification of isolated algebraic singularities by their Alexander polynomials, Topology 23 (1984) 277–287.
- [39] O. Zariski, General theory of saturation and saturated local rings II, Amer. J. Math. 93 (1971) 872–964.